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A randomized controlled trial on the effectiveness of low-level laser therapy versus paracetamol-caffeine for pain control during overall orthodontic treatment

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This study aimed to evaluate the analgesic effects of low-level laser therapy (LLLT) and paracetamol-caffeine in controlling orthodontic pain induced during different stages of leveling and alignment orthodontic treatment. Fifty-four patients with mild to moderate crowded arches were enrolled. Elastomeric separators were placed, followed by fixed appliance bonding one week later. Archwires were replaced according to a predefined sequence until reaching the final 0.019 × 0.025-inch stainless steel wires. The first group received a beam of GaAlAs-Laser before separator placement or any archwire replacement, whereas the second group received paracetamol-caffeine tablets (the drug). Patients in the control group did not receive any procedure. A numeric rating scale (NRS) was used to assess spontaneous and chewing pain immediately, after 1, 24, 48 h, one week, and on the next visit of any replacement. The pain perception reached its peak after 24 h of any replacement. The intensity of pain after 24 and 48 h of separators' placement and initial archwires' engagement was significantly greater than pain induced by 0.016, 0.016 × 0.022, 0.017 × 0.025, and 0.019 × 0.025 NiTi and 0.019 × 0.025 SS archwires in all groups for spontaneous and chewing pain scores. Pain scores in the LLLT group were significantly smaller than those of the control group after 48 h of separation and after 24 and 48 h of rectangular archwire insertion. No significant differences were observed between the drug and control groups. The highest pain levels were induced during the initial stages (separation and initial archwire) of orthodontic treatment. LLLT was able to reduce the peak of high-level pain just during separation and was not highly effective throughout the treatment course. On the other hand, the paracetamol-caffeine combination was not very effective in reducing pain perception throughout the treatment course.

This trial was registered at Clinical Trials.gov (Identifier NCT03400111), registered on 17/01/2018.

Keywords Orthodontic pain, Low level laser therapy, LLLT, Paracetamol-caffeine, Orthodontic archwire

Orthodontic treatment is commonly associated with pain as an adverse effect, with a high incidence of up to 90%[1]. Pain and discomfort may arise following any procedure during orthodontic treatment, including the orthodontic separators placement[2], archwires replacement[3,4], canine retraction[5], and debonding of the braces[6]. Additionally, concerns about pain may negatively impact oral hygiene[7].

Nonsteroidal anti-inflammatory drugs (NSAIDs) are mainly used for controlling orthodontic pain. Commonly used NSAIDs include ibuprofen, ketoprofen, and meloxicam[8–10]. Due to their mechanism of

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action, NSAIDs may have an adverse effect on the orthodontic tooth movement rate especially on the long-term consumption[11]. NSAIDs, block the formation of prostaglandins (PGE2) by inhibiting cyclooxygenase. Therefore, additional alternatives have been proposed to enhance orthodontic pain management. Benzocaine adhesive patches or gels were also used for this purpose[10]. Benzocaine has the potential to numb taste buds, and gels can be spread over a large area in the mouth[2]. Paracetamol is an over-the-counter (OTC) medication that can be used safely to alleviate orthodontic pain. It works by inhibiting the third isoform of the cyclooxygenase enzyme (COX3), which is mainly present in the heart and cerebral cortex[8]. Numerous studies have examined the effectiveness of paracetamol in managing orthodontic pain and have shown that it is not significantly different from placebo[9]. Recently, many pharmacologic manufactures produce a combination paracetamol and caffeine claiming its superiority to paracetamol alone, as caffeine is commonly used as an analgesic drug adjuvant[12]. Its mechanism of action may be related to its ability to alternate the pharmacokinetics of paracetamol[13] and may also produce alterations in mood[14].

The other non-pharmacological methods include vibrational forces, chewing gum[15], bite wafers[16], transcutaneous electrical nerve stimulation, behavioral interventions, static magnetic fields, molecular targeted therapy[1], and low-level laser therapy (LLLT)[2,17,18].

Some trials have concluded that low-level laser therapy is ineffective for controlling orthodontic pain[7,19]. On the other side, others reported that LLLT is an effective intervention for reducing the perception of orthodontic pain[4,18,20–22].

The control of pain perception was frequently evaluated following the use of orthodontic separators[2,8,9,18,20–24]. Some orthodontic studies assessed the levels of pain and discomfort after the initial archwire engagement[4,7,15,16,25,26]. The processes of orthodontic separation and initial archwire installation induce pain that starts immediately after placement with a peak level (moderate to high) at day 2 that might last for 7 days[2,4,27]. However, almost all studies in the field of pain management during orthodontic treatment have just focused on only one stage of the orthodontic treatment. Some trials have tested the perception of pain during canine retraction movement and reported a low level of pain peaked during the first day of retraction[5], others reported a small scores of pain following the final archwire replacement[28] or following the rapid or slow maxillary expansion[29].

Considering the limited availability of comprehensive studies on pain perception throughout the orthodontic treatment stages, ongoing debates about LLLT's efficacy in pain reduction, orthodontists' preference for safer alternatives to NSAIDs, and the absence of research on paracetamol-caffeine for orthodontic pain relief, this study aims to assess the effectiveness of LLLT and paracetamol-caffeine in managing orthodontic pain throughout treatment stages and evaluate pain levels at each stage. The null hypothesis was neither LLLT nor paracetamol-caffeine were effective in reducing orthodontic pain induced throughout the entire orthodontic treatment stages.

Materials and methods

Study design and settings

This trial was a three-arm parallel-group randomized controlled clinical trial. The current study was carried out at the Department of Orthodontic and Dentofacial Orthopedic and the Unit of Laser Research at the Faculty of Dentistry, Damascus University, between February 2018 and January 2020, and approved by the Local Research Ethics Committee at the University of Damascus (UDDS-1199-15082017/SRC-2311). All methods were carried out in accordance with relevant guidelines and regulations. This trial was registered at Clinical Trials.gov (Identifier: NCT03400111), on 17/01/2018. The protocol of the current study can be assessed on Clinical Trials.gov.

Sample size calculation

The sample size was determined using Minitab 18.1 (INC, State College, PA), through the following assumptions: the minimally important clinical difference to be detected was 2 points on a numeric rating scale (NRS)[30], the standard deviation was 2 points in a previous study[7], the level of significance (alpha value) was set as 0.05, with a power of 80%, and employing one-way ANOVA to test the differences among the three groups. The required sample size was 18 patients for each group.

Participants and eligibility criteria

112 patients who had Class I molars and canines relationship according to Angle's classification malocclusion were assessed for eligibility, 51 of them were excluded, and 61 patients met the inclusion criteria of this study. Information sheets were provided, and informed consent was obtained from 57 patients; 4 patients rejected to get involved. 54 subjects were selected with a simple random sampling method (22 male, 32 female). All patients were included in this trial had mild to moderate upper and lower crowded dental arches (tooth-size-arch-length-discrepancy of 2–4 mm), 16–30 years old, normal overbite/overjet, good proximal contact points between the first molar and the adjacent teeth, good oral hygiene, and absence of any systemic and psychiatric diseases. Patients with any gingival melanin pigmentation, prolonged pain/inflammation medication intake, severe malocclusion in any plane, and who received previous orthodontic treatment were excluded.

Randomization and blinding

The randomization was performed according to a web-based-software generated list of random numbers (www.randomizer.org). The recruited patients were assigned to three groups with a 1:1:1 allocation ratio. This process was carried out by one of the academic staff who was not involved in this research project. Allocation concealment was achieved using sequentially numbered, opaque, sealed envelopes which were opened just before the start of the treatment. The outcome assessor was blinded during the data analysis. The principal researcher and patients were not blinded.

Intervention and control groups

All patients received a non-extraction orthodontic treatment to resolve the crowding problem. Firstly, orthodontic separators were placed for one week proximal to the upper and lower first molars on both sides. After 1 week, the first molars were banded (TRUFIT2.0, Orthotechnolgy, Florida, USA) and preadjusted edgewise 0.022-inch (in) slot MBT prescription brackets (Votion, Orthotechnolgy, Florida, USA) were bonded for the upper and lower arches in the same visit.

Alignment and leveling were started with 0.014- in super-elastic nickel-titanium (NiTi) wire followed by 0.016, 0.016*0.022, 0.017*0.025, 0.019*0.025-in NiTi wires and the final archwire was 0.019*0.025 stainless steel (SS), changed at 3-week intervals between each wire. The replacement of archwires was based on a careful evaluation of teeth alignment progress, ensuring that the subsequent archwire could be smoothly inserted with minimal bending, without placing excessive pressure on the anterior teeth or causing difficulties in full ligation[31].

The initial archwire was engaged a week after fixed appliance bonding, to avoid any interference between the pain perception resulting from the mucosal membrane irritation and the pain induced by the initial wire. Archwires were engaged using elastic ligature and cut distal to the first molar tubes, in such a way that they did not hurt the buccal mucosa.

Laser irradiation group

Before the placement of orthodontic separators and any archwire replacement, patients in the LLLT group were irradiated with a continuous infrared beam of gallium-aluminum-arsenide (Ga-Al-Ar) semiconductor laser device (Konf, konftec, Klas-DX84, New Taipei, Taiwan) which may has more depth of penetration in the irradiated tissue[4]. The diameter of laser tip, 8 mm (2.01 cm^2); the wavelength, 808 nanometers (nm); the power 350 milliWatt (mW); 10 s, of irradiation per point; and 3.5 J/point for each application point using continuous mode. The irradiation dose per tooth was 3.49 J/cm^2.

A total of 12 teeth in the upper and lower arches at the same session, from the left first molar to the right first molar, were irradiated at the center of the root on both the buccal and palatal sides of each tooth (Fig. 1). The laser device tip was directed almost perpendicularly to the irradiated surface, and tightly touched the attached gingiva or the oral mucosa. Cotton rolls and a water-air syringe were used to ensure good isolation. Protective glasses were used by the operator and the patient. The irradiation dose in a single arch was 41.88 J/cm2, and the overall irradiation dose was 83.76 J/cm^2.

Paracetamol–caffeine (the drug) group

Patients in the medication group were instructed to take paracetamol–caffeine 500–65 mg (Panadol Extra, GlaxoSmithKline, KSA) three times daily (one tablet every 8 h) during the first two days following separator insertion or archwire replacement, and thereafter as needed. To ensure that the analgesic action of the drug was taking effect before any replacement, patients were asked to take the tablet one hour before any appointment. All patients were monitored by the principal researcher to ensure that they took their medications and filled their questionnaire in appropriate time.

Control group

Patients in the control group did not receive any intervention.

Outcomes measures

The spontaneous and chewing pain scores were evaluated with a questionnaire was formulated using an 11-point numeric rating scale (NRS), with 0 indicating "no pain" and 10 indicating "extreme pain". The score was assessed immediately following separation or after any archwire replacement, and the the following assessment points: at 1 h, 24 h, 48 h, 1 week, and on the next visit[2].

In certain cases, archwire replacement could not be performed at the 3-week interval. In such situations, the placed archwire was maintained for an additional 3 weeks before proceeding with the next archwire, following the criteria described above. Pain assessment in these cases was conducted during the initial 3-week period.

Pain responses were recorded using paper-based questionnaires distributed to patients at each follow-up appointment. Using SMS or WhatsApp messages, all patients were prompted to fill out the questionnaires at the scheduled times. In case of unbearable pain levels, patients were directed to contact the principal investigator in order to get his advice which was: "Please wait to fill in the questionnaire before taking any medication".

Statistical analysis

The statistical package for the social sciences (SPSS) software, version 20 (IBM ®, Armonk, NY, USA) was used to perform the statistical analyses. The normality of the data distribution was tested with Shapiro–Wilk test. The Kruskal– Wallis test was used to detect the differences in the spontaneous and chewing pain scores. For pairwise comparisons, Mann– Whitney U tests were used as a post hoc test. When the data were normally distributed, the one-way ANOVA test was used with Tukey's honestly significant difference (HSD) post-hoc tests for pairwise comparisons.

Results

Participant flow and follow-up

One hundred and twelve class I malocclusion patients were examined for eligibility; 51 were excluded, and 61 met the inclusion criteria of the current trial. A total of 54 patients were randomly selected from the sampling frame and randomly allocated to one of the three groups. The details of enrolment, intervention allocation, follow-up, and data analysis are described in the CONSORT flow diagram (Fig. 2). The sample basic characteristics are given in Table 1.

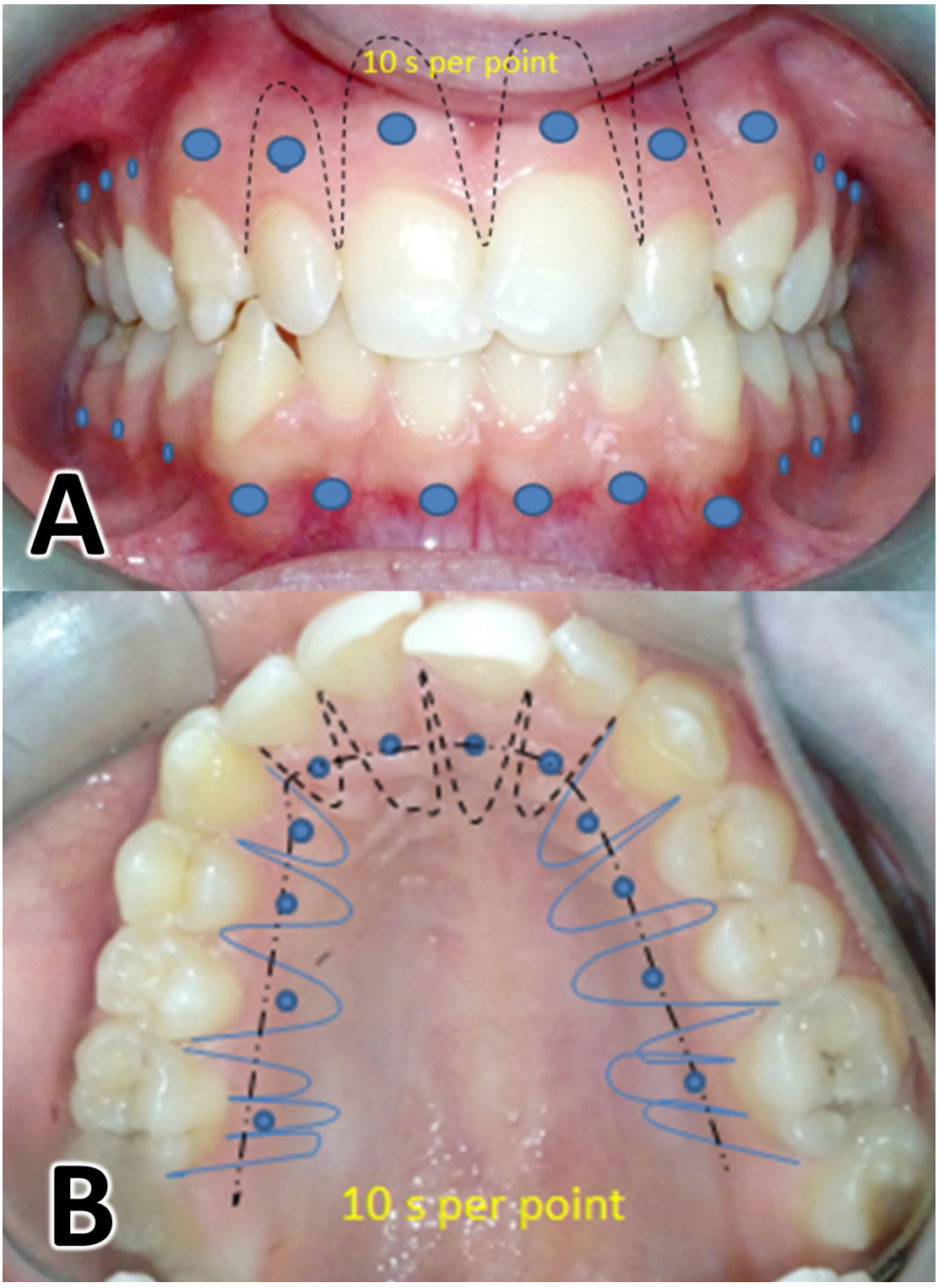


Fig. 1. The irradiated points at the center of the root on both the buccal and palatal sides of each tooth in the upper and lower arches. A: frontal view, B: occlusal view.

In the current study, the pain perception was analyzed for each treatment stage independently. Most of the patients reported that the pain peaked after 24 h of any replacement and persisted for 2 days and began to disappear thereafter. The chewing pain mean scores were greater than those of spontaneous pain mean scores during the separation and all stages of alignment and leveling. In all groups, the maximum intensity of spontaneous and chewing pain ranged from low to moderate levels as shown in Tables 2, 4 and 3).

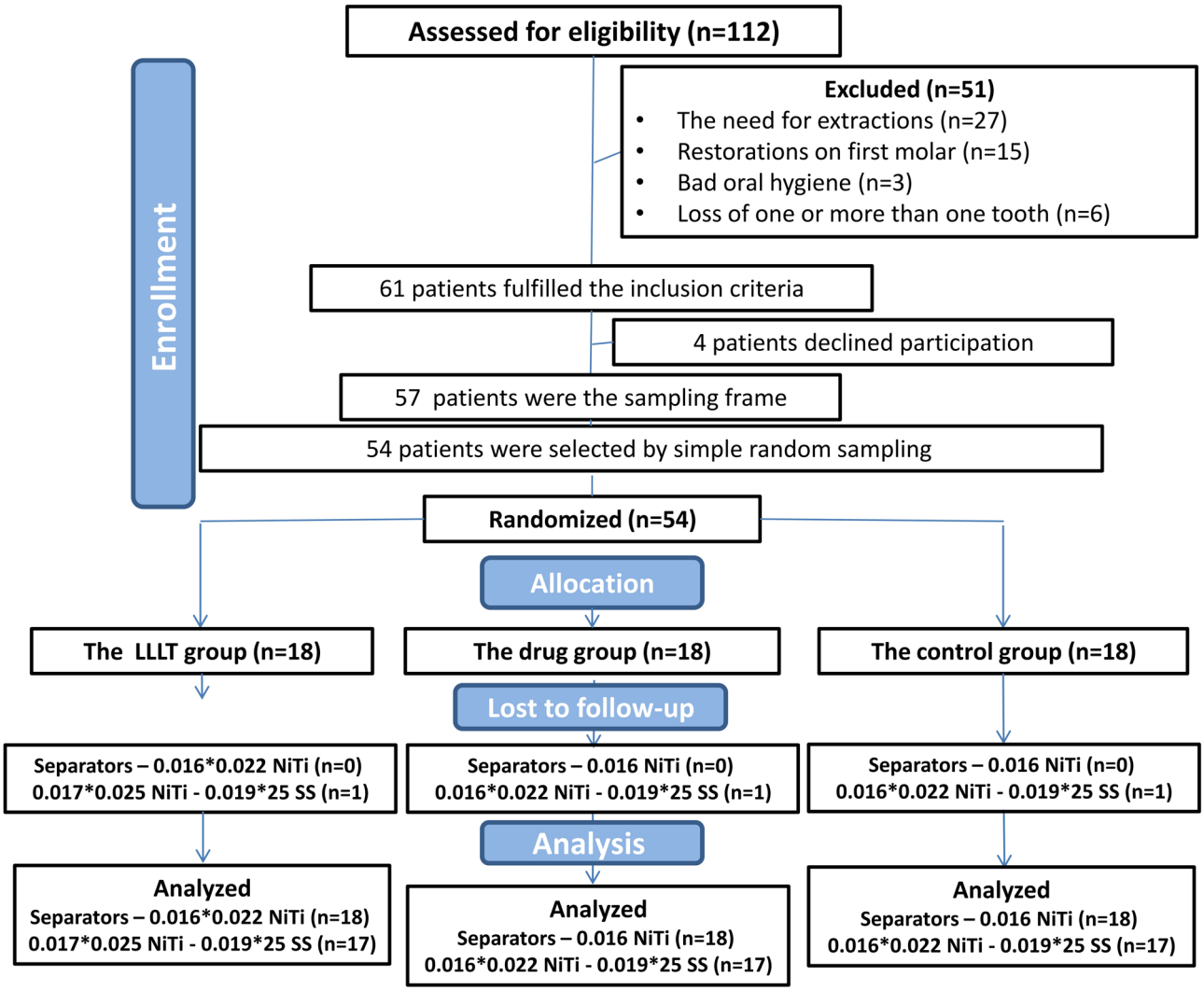


Fig. 2. CONSORT flow diagram of patients' recruitment, follow-up, and entry into data analysis.

Group	Gender			Age mean (± S.D)
	N	Female	Male	
Laser group	18	11	7	21.11 (± 2.08)
Drug group	18	11	7	21.94 (± 3.52)
Control group	18	10	8	22.00 (± 2.58)
Total	54	32	22	21.68 (± 2.77)

Table 1. Baseline sample characteristics. N: sample size; S.D: standard deviation.

During the separation stage, the spontaneous and chewing pain mean scores in the LLLT group were significantly smaller than those of the control group after 1 h ($P = 0.002$ and $p = 0.003$ for spontaneous and chewing scores respectively) and 48 h ($p = 0.014$ and $p = 0.007$ for spontaneous and chewing scores respectively) of separation. On the other hand, pain perception was similar between the three groups immediately, after 24 h and 1 week of separators placement ($P > 0.05$) as shown in Table II. For the spontaneous and chewing pain perception, the levels of pain were similar between the 3 groups in all assessment times after the initial (0.014-inch) and the second (0.016-inch) archwire insertion with no significant differences; Table 2.

After the insertion of rectangular archwires (0.016*0.022 NiTi, 0.017*0.025 NiTi, 0.019*0.025 NiTi, and 0.019*0.025 SS), the mean score of pain among patients in the LLLT group was significantly lower than that in the control group after 24 h and 48 h ($p < 0.05$). On the other hand, no statistically significant differences were observed between the LLLT, drug, and control groups at the other assessment times as mentioned in Tables 4 and 3.

Separators										
			Spontaneous pain				Chewing pain			
					Pairwise comparisons §				Pairwise comparisons§	
Time	Group	N	Mean (SD)	P-value*	Pair	P-value†	Mean (SD)	P-value*	Pair	P-value†
h0	Laser	18	0.83 (1.04)	0.430			0.50 (1.20)	0.054		
	Drug	18	1.27 (1.31)				1.72 (1.93)			
	Control	18	1.38 (1.46)				1.83 (1.54)			
1 h	Laser	18	0.44 (0.85)	0.005	Laser vs. Drug	0.063	0.77 (1.21)	0.011	Laser vs. Drug	0.051
	Drug	18	1.88 (2.63)		Laser vs. Control	0.002	2.55 (3.01)		Laser vs. Control	0.003
	Control	18	2.38 (2.37)		Drug vs. Control	0.352	2.94 (2.57)		Drug vs. Control	0.411
24 h	Laser	18	3.77 (2.75)	0.086			5.22 (2.79)	0.281‡		
	Drug	18	3.61 (2.83)				5.94 (2.66)			
	Control	18	5.22 (2.51)				6.66 (2.58)			
48 h	Laser	18	3.33 (2.22)	0.044	Laser vs. Drug	0.323	4.83 (2.22)	0.010‡	Laser vs. Drug	*0.263*
	Drug	18	3.61 (1.68)		Laser vs. Control	0.014	5.94 (1.92)		Laser vs. Control	*0.007*
	Control	18	5.11 (2.58)		Drug vs. Control	0.068	7.05 (2.15)		Drug vs. Control	*0.263*
1 week	Laser	18	0.83 (1.20)	0.276			2.05 (2.01)	0.114		
	Drug	18	1.16 (1.50)				2.44 (1.91)			
	Control	18	1.66 (2.32)				3.55 (2.63)			

0.014 NiTi and 0.016 NiTi archwires										
			0.014 inch archwire				0.016 inch archwire			
			Spontaneous pain		Chewing pain		Spontaneous pain		Chewing pain	
Time	Group	*N*	Mean (SD)	*P*-value*	Mean (SD)	*P*-value*	Mean (SD)	*P*-value*	Mean (SD)	*P*-value*
h0	Laser	18	0.56 (0.92)	**0.135**	1.61 (2.43)	**0.700**	0.67 (1.24)	**0.056**	1 (1.08)	**0.140**
	Drug	18	1.22 (1.56)		1.67 (2.03)		1.28 (0.89)		1.83 (1.50)	
	Control	18	1.33 (1.78)		1.61 (1.79)		0.94 (1.21)		1.17 (1.38)	
1 h	Laser	18	1 (1.33)	**0.079**	1.78 (2.34)	0.124	1.11 (1.68)	**0.422**	1.61 (1.72)	0.409
	Drug	18	2 (2.28)		2.89 (2.42)		1.56 (1.65)		2.28 (2.02)	
	Control	18	2.28 (2.14)		3.44 (2.89)		1.72 (1.99)		2.33 (2.11)	
24 h	Laser	18	2.83 (2.50)	**0.322**	4.56 (2.45)	**0.308** ǂ	0.89 (1.57)	**0.067**	1.78 (1.52)	**0.299**
	Drug	18	3.50 (2.66)		6 (3.11)		1.44 (1.76)		2.11 (1.71)	
	Control	18	3.94 (2.96)		5.44 (2.85)		2.17 (2.36)		2.78 (2.29)	
48 h	Laser	18	2.78 (3.32)	**0.564**	4.11 (2.65)	0.522	0.83 (1.42)	**0.192**	1.44 (1.46)	0.160
	Drug	18	3.11 (2.68)		5.22 (3.28)		1.33 (1.64)		1.33 (1.19)	
	Control	18	3.33 (2.97)		4.89 (3.03)		1.83 (2.28)		2.50 (2.43)	
1 week	Laser	18	0.94 (1.30)	**0.360**	1.89 (1.71)	**0.212**	0.44 (0.92)	**0.264**	0.78 (0.88)	**0.260**
	Drug	18	1.44 (2.20)		3.22 (2.88)		0.83 (1.62)		0.44 (0.62)	
	Control	18	2.06 (2.34)		3.61 (3.15)		1 (1.28)		1.56 (1.50)	
Next visit	Laser	18	0.11 (0.47)	**0.160**	0.56 (0.98)	0.320	0.11 (0.32)	**0.308**	0.11 (0.32)	0.242
	Drug	18	0.67 (1.03)		0.89 (1.84)		0.11 (0.32)		0.39 (0.70)	
	Control	18	1.06 (1.35)		1.94 (2.29)		0.28 (0.46)		0.44 (0.70)	

Table 2. Descriptive statistics of levels of spontaneous and chewing pain during separation and the early stages of leveling and alignment in the three groups along with p-values of significance testing. * Kruskal-Wallis test was applied unless another test was indicated; †: Mann-Whitney U test was used for pairwise comparisons unless another test was indicated; ǂ: One-way ANOVA; underlined P-values are related to Tukey's HSD tests; §: pairwise comparisons were applied when a significant difference was found between the groups in general; level of significance was set at 0.05 and Bonferroni's correction was applied for post-hoc tests (i.e., adjusted $\alpha = 0.016$); N: number of patients; h0: immediately.

At both the 24-hour and 48-hour intervals following separator placement, the intensity of spontaneous pain (mean values: 3.77, 3.61, 5.22 after 24 h and 3.33, 3.61, 5.11 after 48 h for the laser, drug, and control, respectively) was notably greater compared to the pain induced by 0.016, 0.016*0.022, 0.017*0.025, 0.019*0.025 NiTi and 0.019*0.025 SS archwires in the LLLT, drug, and control groups. ($p < 0.001$). Furthermore, the initial archwire-induced pain mean scores were greater than those of 0.016, 0.017*0.025, and 0.019*0.025 SS in the 3 groups as mentioned in Table 5.

As shown in Table 6, the same findings were repeated with the chewing pain perception. No significant differences were observed between the treatment stages immediately, after 1 h, a week, and on the next visit

0.019*0.025 NiTi										
			Spontaneous pain				Chewing pain			
					Pairwise comparisons §				Pairwise comparisons§	
Time	Group	N	Mean (SD)	P-value*	Pair	P-value†	Mean (SD)	P-value*	Pair	P-value†
h0	Laser	17	0.6 (0.6)	0.499			0.8 (0.8)	0.333		
	Drug	17	0.8 (0.9)				0.8 (1)			
	Control	17	1.2 (1.4)				1.5 (1.4)			
1 h	Laser	17	0.9 (1)	0.138			1.4 (0.9)	0.183		
	Drug	17	0.9 (1.1)				1.4 (1.7)			
	Control	17	1.8 (1.5)				2.2 (1.8)			
24 h	Laser	17	1.6 (1.5)	0.007	Laser vs. Drug	0.049	2.5 (1.9)	0.045	Laser vs. Drug	0.062
	Drug	17	0.7 (1.2)		Laser vs. Control	0.005	1.4 (1.8)		Laser vs. Control	0.012
	Control	17	2.8 (2.2)		Drug vs. Control	0.131	3.1 (2.2)		Drug vs. Control	0.586
48 h	Laser	17	1.5 (1.5)	0.019	Laser vs. Drug	0.053	2.1 (2)	0.042	Laser vs. Drug	0.073
	Drug	17	0.5 (1)		Laser vs. Control	0.015	0.9 (1.4)		Laser vs. Control	0.015
	Control	17	2.5 (2.5)		Drug vs. Control	0.322	2.7 (2.3)		Drug vs. Control	0.496
1 week	Laser	17	0.2 (0.5)	0.056			0.5 (0.8)	0.192		
	Drug	17	0.2 (0.6)				0.4 (0.7)			
	Control	17	1.1 (1.4)				1.2 (1.5)			
Next visit	Laser	17	0.1 (0.2)	0.128			0.1 (0.5)	0.307		
	Drug	17	0.1 (0.3)				0.2 (0.5)			
	Control	17	0.5 (0.9)				0.5 (0.9)			
0.019*0.025 SS										
			Spontaneous pain				Chewing pain			
					Pairwise comparisons §				Pairwise comparisons§	
Time	Group	*N*	Mean (SD)	*P*-value*	Pair	*P*-value†	Mean (SD)	*P*-value*	Pair	*P*-value†
h0	Laser	17	0.4 (0.6)	0.055			0.5 (0.8)	0.390		
	Drug	17	0.7 (0.7)				1.1 (1.1)			
	Control	17	1.2 (1.2)				1.4 (1.2)			
1 h	Laser	17	0.7 (0.8)	0.260			0.8 (0.7)	0.090		
	Drug	17	0.9 (0.9)				1.5 (1.3)			
	Control	17	1.7 (1.2)				2.2 (1.3)			
24 h	Laser	17	0.5 (0.7)	0.007	Laser vs. Drug	0.218	0.9 (1)	0.015	Laser vs. Drug	0.182
	Drug	17	1.1 (1.1)		Laser vs. Control	0.003	1.6 (1.3)		Laser vs. Control	0.006
	Control	17	2.4 (2.3)		Drug vs. Control	0.053	2.9 (2.4)		Drug vs. Control	0.099
48 h	Laser	17	0.5 (0.9)	0.021	Laser vs. Drug	0.865	0.8 (0.8)	0.042	Laser vs. Drug	0.786
	Drug	17	0.5 (0.9)		Laser vs. Control	0.014	0.9 (1.3)		Laser vs. Control	0.007
	Control	17	1.9 (1.9)		Drug vs. Control	0.034	2.3 (2.1)		Drug vs. Control	0.231
1 week	Laser	17	0.4 (0.8)	0.330			0.2 (0.6)	0.420		
	Drug	17	0.2 (0.7)				0.5 (0.9)			
	Control	17	1 (1.4)				1.3 (1.6)			
Next visit	Laser	17	0.0 (0)	0.190			0.1 (0.2)	0.060		
	Drug	17	0.1 (0.2)				0.1 (0.2)			
	Control	17	0.7 (1.7)				0.8 (1.8)			

Table 3. Descriptive statistics of levels of spontaneous and chewing pain during 0.019*0.025 NiTi and 0.019*0.025 SS archwires in the three groups along with p-values of significance testing. * Kruskal-Wallis test was applied; †: Mann-Whitney U test was used for pairwise comparisons; §: pairwise comparisons were applied when a significant difference was found between the groups in general; level of significance was set at 0.05 and Bonferroni's correction was applied for post-hoc tests (i.e., adjusted $\alpha = 0.016$); N: number of patients; h0: immediately after activation.

in the LLLT, drug, and control groups ($p > 0.05$). The greatest pain perception occurred during chewing at the separation stage (mean values: 5.22, 5.94, 6.66 after 24 h and 4.83, 5.94, 7.05 after 48 h for the laser, drug, and control respectively), followed by initial archwire-induced chewing pain (mean values: 4.56, 6, 5.44 after 24 h and 4.11, 5.22, 4.89 after 48 h for the laser, drug, and control respectively). 95% confidence intervals of means for spontaneous and chewing scores are listed in Supplementary Table 1.

0.016*0.022 inch NiTi archwire										
			Spontaneous pain				Chewing pain			
			Mean (SD)	P-value*	Pairwise comparisons §				Pairwise comparisons§	
Time	Group	N			Pair	P-value†	Mean (SD)	P-value*	Pair	P-value†
h0	Laser	17	**0.7 (0.8)**				**0.9 (0.9)**			
	Drug	17	**1.1 (0.8)**	**0.327**			**1.2 (1.2)**	**0.550**		
	Control	18	**1.5 (2.0)**				**1.8 (2.1)**			
1 h	Laser	17	**1.4 (1.6)**				**1.8 (1.6)**			
	Drug	17	**1.2 (0.9)**	**0.188**			**1.8 (1.1)**	0.177		
	Control	18	**2.4 (2.3)**				**3.2 (2.7)**			
24 h	Laser	17	**2.2 (2.5)**		Laser vs. Drug	0.245	**3.1 (2.7)**		Laser vs. Drug	0.231
	Drug	17	**1.1 (0.9)**	**0.017**	Laser vs. Control	0.003	**1.9 (1.2)**	**0.033**	Laser vs. Control	0.007
	Control	18	**3.1 (2.2)**		Drug vs. Control	0.184	**3.8 (2.2)**		Drug vs. Control	0.273
48 h	Laser	17	**2.0 (2.6)**		Laser vs. Drug	0.045	**2.7 (3)**		Laser vs. Drug	0.150
	Drug	17	**0.5 (0.8)**	**0.003**	Laser vs. Control	0.001	**1.1 (1.1)**	0.010	Laser vs. Control	0.001
	Control	18	**2.7 (2.4)**		Drug vs. Control	0.207	**3.4 (2.4)**		Drug vs. Control	0.232
1 week	Laser	17	**0.9 (1.2)**				**1.8 (1.6)**			
	Drug	17	**0.4 (0.6)**	**0.349**			**0.9 (1.2)**	**0.254**		
	Control	18	**1.1 (1.9)**				**1.8 (2.2)**			
Next visit	Laser	17	**0.0 (0.0)**				**0.2 (0.4)**			
	Drug	17	**0.2 (0.4)**	**0.130**			**0.4 (0.6)**	0.486		
	Control	18	**0.8 (1.5)**				**0.8 (1.8)**			
0.017*0.025 inch NiTi archwire										
			Spontaneous pain				**Chewing pain**			
					Pairwise comparisons §				**Pairwise comparisons§**	
Time	**Group**	***N***	**Mean (SD)**	***P*-value***	**Pair**	***P*-value†**	**Mean (SD)**	***P*-value***	**Pair**	***P*-value†**
h0	Laser	17	0.4 (0.5)				0.6 (0.9)			
	Drug	17	0.9 (1)	**0.161**			0.9 (1.4)	0.183		
	Control	17	1.2 (1.6)				1.6 (1.8)			
1 h	Laser	17	0.4 (1)				0.8 (1)			
	Drug	17	0.9 (1)	**0.150**			1.2 (1.3)	0.140		
	Control	17	1.7 (1.8)				2.1 (1.4)			
24 h	Laser	17	0.6 (1.1)		Laser vs. Drug	0.131	0.8 (1.1)		Laser vs. Drug	0.357
	Drug	17	1.3 (1.4)	**0.017**	Laser vs. Control	0.005	1.5 (1.9)	0.004	Laser vs. Control	0.001
	Control	17	1.9 (1.6)		Drug vs. Control	0.259	2.4 (1.3)		Drug vs. Control	0.038
48 h	Laser	17	0.4 (0.8)		Laser vs. Drug	1	0.8 (1.2)		Laser vs. Drug	0.634
	Drug	17	0.4 (0.6)	**0.006**	Laser vs. Control	0.013	0.5 (0.8)	0.009	Laser vs. Control	0.005
	Control	17	1.3 (1.4)		Drug vs. Control	0.120	1.9 (1.5)		Drug vs. Control	0.038
1 week	Laser	17	0.4 (0.9)				0.6 (0.9)			
	Drug	17	0.1 (0.3)	**0.106**			0.2 (0.4)	0.170		
	Control	17	0.8 (1.6)				1.1 (1.3)			
Next visit	Laser	17	0.0 (0)				0.0 (0)			
	Drug	17	0.1 (0.2)	**0.364**			0.1 (0.2)	0.151		
	Control	17	0.2 (0.5)				0.2 (0.6)			

Table 4. Descriptive statistics of levels of spontaneous and chewing pain during 0.016*0.022 NiTi and 0.017*0.025 NiTi archwires in the three groups along with p-values of significance testing. * Kruskal-Wallis test was applied; †: Mann-Whitney U test was used for pairwise comparisons; §: pairwise comparisons were applied when a significant difference was found between the groups in general; level of significance was set at 0.05 and Bonferroni's correction was applied for post-hoc tests (i.e., adjusted $\alpha = 0.016$); N: number of patients; h0: immediately after activation.

Discussion

In the current trial, the effectiveness of a single dose of the low-level laser before each stage of the orthodontic leveling and alignment treatment and paracetamol– caffeine combination drug in reliving the orthodontic pain were evaluated.

The current research work was a three-arm, parallel-design study. Some researchers have claimed that the split-mouth design is superior to the parallel design in the trials that evaluate patient-centered outcomes to overcome

		Separators Mean (SD)	0.014 NITI Mean (SD)	0.016 NITI Mean (SD)	0.016*0.022 NITI Mean (SD)	0.017*0.025 NITI Mean (SD)	0.019*0.025 NITI Mean (SD)	0.019*0.025 SS Mean (SD)	*p*-value*
Laser group	**H0**	0.83 (1.04)	0.56 (0.92)	0.67 (1.24)	0.7 (0.8)	0.4 (0.5)	0.6 (0.6)	0.4 (0.6)	0.609
	1 h	0.44 (0.85)	1 (1.33)	1.11 (1.68)	1.4 (1.6)	0.4 (1)	0.9 (1)	0.7 (0.8)	0.286
	24 h	3.77 (2.75)	2.83 (2.50)	0.89 (1.57)	2.2 (2.5)	0.6 (1.1)	1.6 (1.5)	0.5 (0.7)	< 0.001
	48 h	3.33 (2.22)	2.78 (3.32)	0.83 (1.42)	2.0 (2.6)	0.4 (0.8)	1.5 (1.5)	0.5 (0.9)	< 0.001
	1 week	0.83 (1.20)	0.94 (1.30)	0.44 (0.92)	0.9 (1.2)	0.4 (0.9)	0.2 (0.5)	0.4 (0.8)	0.080
	Next visit		0.11 (0.47)	0.11 (0.32)	0.0 (0.0)	0.0 (0)	0.1 (0.2)	0.0 (0)	0.098
Drug group	**H0**	1.27 (1.31)	1.22 (1.56)	1.28 (0.89)	1.1 (0.8)	0.9 (1)	0.8 (0.9)	0.7 (0.7)	0.534
	1 h	1.88 (2.63)	2 (2.28)	1.56 (1.65)	1.2 (0.9)	0.9 (1)	0.9 (1.1)	0.9 (0.9)	0.461
	24 h	3.61 (2.83)	3.50 (2.66)	1.44 (1.76)	1.1 (0.9)	1.3 (1.4)	0.7 (1.2)	1.1 (1.1)	< 0.001
	48 h	3.61 (1.68)	3.11 (2.68)	1.33 (1.64)	0.5 (0.8)	0.4 (0.6)	0.5 (1)	0.5 (0.9)	< 0.001
	1 week	1.16 (1.50)	1.44 (2.20)	0.83 (1.62)	0.4 (0.6)	0.1 (0.3)	0.2 (0.6)	0.2 (0.7)	0.122
	Next visit		0.67 (1.03)	0.11 (0.32)	0.2 (0.4)	0.1 (0.2)	0.1 (0.3)	0.1 (0.2)	0.190
Control group	**H0**	1.38 (1.46)	1.33 (1.78)	0.94 (1.21)	1.5 (2.0)	1.2 (1.6)	1.2 (1.4)	1.2 (1.2)	0.972
	1 h	2.38 (2.37)	2.28 (2.14)	1.72 (1.99)	2.4 (2.3)	1.7 (1.8)	1.8 (1.5)	1.7 (1.2)	0.915
	24 h	5.22 (2.51)	3.94 (2.96)	2.17 (2.36)	3.1 (2.2)	1.9 (1.6)	2.8 (2.2)	2.4 (2.3)	< 0.001
	48 h	5.11 (2.58)	3.33 (2.97)	1.83 (2.28)	2.7 (2.4)	1.3 (1.4)	2.5 (2.5)	1.9 (1.9)	< 0.001
	1 week	1.66 (2.32)	2.06 (2.34)	1 (1.28)	1.1 (1.9)	0.8 (1.6)	1.1 (1.4)	1 (1.4)	0.369
	Next visit		1.06 (1.35)	0.28 (0.46)	0.8 (1.5)	0.2 (0.5)	0.5 (0.9)	0.7 (1.7)	0.051

***p*-values for multiple comparisons (post hoc tests) §**						
	Laser group		**Drug group**		**Control group**	
Pair	**24 h †**	**48 h †**	**24 h †**	**48 h †**	**24 h †**	**48 h †**
SP vs. 0.014	0.239	0.051	0.963	0.171	0.111	0.037
SP vs. 0.016	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
SP vs. 16.22	< 0.001	< 0.001	0.001	< 0.001	0.001	0.001
SP vs. 17*25	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
SP vs. 19*25	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	0.001
SP vs. 19.25SS	< 0.001	< 0.001	0.001	< 0.001	0.001	< 0.001
0.014 vs. 0.016	< 0.001	< 0.001	0.001	< 0.001	0.001	0.001
0.014 vs. 16*22	0.245	0.351	0.020	0.010	0.339	0.606
0.014 vs. 17*25	< 0.001	< 0.001	0.001	< 0.001	0.001	0.001
0.014 vs. 19*25	0.134	0.483	0.010	0.004	0.245	0.386
0.014 vs. 19*25 ss	< 0.001	< 0.001	0.001	< 0.001	0.001	< 0.001
0.016 vs. 16*22	0.022	0.096	0. 935	0.153	0.214	0.203
0.016 vs. 17*25	0.732	0.636	0.987	0.057	0.935	0.684
0.016 vs. 19*25	0.077	0.163	0.232	0.072	0.273	0.483
0.016 vs. 19*25ss	0.858	0.684	0.782	0.067	0.636	0.832
16*22 vs. 17*25	0.007	0.014	0.973	0.658	0.118	0.045
16*22 vs. 19*25	0.683	0.973	0.114	0.708	0.757	0.757
16*22 vs. 19*25ss	0.008	0.022	0.683	0.683	0.424	0.351
17*25 vs. 19*25	0.029	0.045	0.205	1	0.274	0.339
17*25 vs. 19*25 ss	0.838	0.946	0.734	0.946	0.586	0.634
19.25 vs. 19.25ss	0.026	0.053	0.290	0.892	0.496	0.496

Table 5. Descriptive statistics of levels of spontaneous pain throughout the leveling and alignment treatment course along with p-values of significance testing in the three groups. * Friedman test was applied; †: Wilcoxon signed ranks test was used for pairwise comparisons; §: pairwise comparisons were applied when a significant difference was found between the groups in general; level of significance was set at 0.05 and Bonferroni's correction was applied for post-hoc tests (i.e., adjusted $\alpha = 0.002$); h0: immediately after activation, SP: separators; ss: stainless steel.

interpersonal differences such as sex, age, and psychiatric changes[17]. However, an inherent shortcoming of this design was the possibility of carrying the perception of pain from one side to another, resulting in difficulty in accurately determining pain by patients, especially after archwire replacement[2,32]. On the other hand, some studies did not find any relation between orthodontic pain perception and age or gender[33].

Adult patients with mild to moderate upper and lower crowded arches were included in the current study. Adults can express their feelings more accurately than younger people[34]. The applied orthodontic force depends

		Separators Mean (SD)	0.014 NITI Mean (SD)	0.016 NITI Mean (SD)	0.016*0.022 NITI Mean (SD)	0.017*0.025 NITI Mean (SD)	0.019*0.025 NITI Mean (SD)	0.019*0.025 SS Mean (SD)	*p*-value*
Laser group	**H0**	0.50 (1.20)	1.61 (2.43)	1 (1.08)	0.9 (0.9)	0.6 (0.9)	0.8 (0.8)	0.5 (0.8)	0.194
	1 h	0.77 (1.21)	1.78 (2.34)	1.61 (1.72)	1.8 (1.6)	0.8 (1)	1.4 (0.9)	0.8 (0.7)	0.166
	24 h	5.22 (2.79)	4.56 (2.45)	1.78 (1.52)	3.1 (2.7)	0.8 (1.1)	2.5 (1.9)	0.9 (1)	< 0.001
	48 h	4.83 (2.22)	4.11 (2.65)	1.44 (1.46)	2.7 (3)	0.8 (1.2)	2.1 (2)	0.8 (0.8)	< 0.001
	1 week	2.05 (2.01)	1.89 (1.71)	0.78 (0.88)	1.8 (1.6)	0.6 (0.9)	0.5 (0.8)	0.2 (0.6)	0.062
	Next visit		0.56 (0.98)	0.11 (0.32)	0.2 (0.4)	0.0 (0)	0.1 (0.5)	0.1 (0.2)	0.054
Drug group	**H0**	1.72 (1.93)	1.67 (2.03)	1.83 (1.50)	1.2 (1.2)	0.9 (1.4)	0.8 (1)	1.1 (1.1)	0.302
	1 h	2.55 (3.01)	2.89 (2.42)	2.28 (2.02)	1.8 (1.1)	1.2 (1.3)	1.4 (1.7)	1.5 (1.3)	0.211
	24 h	5.94 (2.66)	6 (3.11)	2.11 (1.71)	1.9 (1.2)	1.5 (1.9)	1.4 (1.8)	1.6 (1.3)	< 0.001
	48 h	5.94 (1.92)	5.22 (3.28)	1.33 (1.19)	1.1 (1.1)	0.5 (0.8)	0.9 (1.4)	0.9 (1.3)	< 0.001
	1 week	2.44 (1.91)	3.22 (2.88)	0.44 (0.62)	0.9 (1.2)	0.2 (0.4)	0.4 (0.7)	0.5 (0.9)	0.540
	Next visit		0.89 (1.84)	0.39 (0.70)	0.4 (0.6)	0.1 (0.2)	0.2 (0.5)	0.1 (0.2)	0.074
Control group	**H0**	1.83 (1.54)	1.61 (1.79)	1.17 (1.38)	1.8 (2.1)	1.6 (1.8)	1.5 (1.4)	1.4 (1.2)	0.908
	1 h	2.94 (2.57)	3.44 (2.89)	2.33 (2.11)	3.2 (2.7)	2.1 (1.4)	2.2 (1.8)	2.2 (1.3)	0.774
	24 h	6.66 (2.58)	5.44 (2.85)	2.78 (2.29)	3.8 (2.2)	2.4 (1.3)	3.1 (2.2)	2.9 (2.4)	< 0.001
	48 h	7.05 (2.15)	4.89 (3.03)	2.50 (2.43)	3.4 (2.4)	1.9 (1.5)	2.7 (2.3)	2.3 (2.1)	< 0.001
	1 week	3.55 (2.63)	3.61 (3.15)	1.56 (1.50)	1.8 (2.2)	1.1 (1.3)	1.2 (1.5)	1.3 (1.6)	0.100
	Next visit		1.94 (2.29)	0.44 (0.70)	0.8 (1.8)	0.2 (0.6)	0.5 (0.9)	0.8 (1.8)	0.087

***p*-values for multiple comparisons (post hoc tests) §**						
	Laser group		**Drug group**		**Control group**	
Pair	**24 h †**	**48 h †**	**24 h †**	**48 h †**	**24 h †**	**48 h †**
SP vs. 0.014	0.488	0.174	0.897	0.406	0.176	0.026
SP vs. 0.016	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
SP vs. 16.22	< 0.001	< 0.001	< 0.001	< 0.001	0.001	< 0.001
SP vs. 17*25	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
SP vs. 19*25	0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
SP vs. 19.25SS	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
0.014 vs. 0.016	< 0.001	< 0.001	< 0.001	< 0.001	0.001	0.001
0.014 vs. 16*22	0.051	0.037	0.004	0.006	0.077	0.165
0.014 vs. 17*25	< 0.001	< 0.001	< 0.001	< 0.001	0.001	0.001
0.014 vs. 19*25	0.014	0.023	0.005	0.004	0.014	0.035
0.014 vs. 19*25 ss	< 0.001	< 0.001	< 0.001	< 0.001	0.001	< 0.001
0.016 vs. 16*22	0.111	0.271	0.800	0.564	0.122	0.140
0.016 vs. 17*25	0.026	0.121	0.204	0.022	0.693	0.682
0.016 vs. 19*25	0.256	0.343	0.168	0.144	0.483	0.699
0.016 vs. 19*25ss	0.078	0.165	0.357	0.112	0.880	1.000
16*22 vs. 17*25	0.002	0.017	0.157	0.090	0.043	0.038
16*22 vs. 19*25	0.648	0.832	0.133	0.364	0.424	0.423
16*22 vs. 19*25ss	0.004	0.027	0.318	0.329	0.211	0.154
17*25 vs. 19*25	0.007	0.042	0.729	0.585	0.329	0.361
17*25 vs. 19*25 ss	0.564	0.666	0.496	0.566	0.623	0.644
19.25 vs. 19.25ss	0.015	0.039	0.250	0.921	0.663	0.583

Table 6. Descriptive statistics of levels of chewing pain throughout the leveling and alignment treatment course along with p-values of significance testing in the three groups. * Friedman test was applied; †: Wilcoxon signed ranks test was used for pairwise comparisons; §: pairwise comparisons were applied when a significant difference was found between the groups in general; level of significance was set at 0.05 and Bonferroni's correction was applied for post-hoc tests (i.e., adjusted $\alpha = 0.002$); h0: immediately after activation, SP: separators; ss: stainless steel.

on the position of the teeth and the crowding level, especially after orthodontic separator placement or initial archwire engagement, as the greater force may cause greater pain[2,35]. Many previous studies did not consider the effect of crowding degree and the selected age group in their study designs[21,22,25].

Similar to many previous studies[2,4,7,18,21,22], a diode Ga-Al-As laser was used with a wavelength of 810 nm (near-infrared electromagnetic spectrum). Infrared radiation has a low absorption coefficient in hemoglobin and water; thus, more depth of penetration in the irradiated tissue, possibly reaching cortical and alveolar bone

tissues[4]. To achieve the best results, it is important to use an appropriate amount of laser energy. In the current study, a power of 3.5 joules per point was applied, similar to the studies of Almallah et al., Owayda et al. after separators placement[2,22] and studies of Qamruddin et al. and Domínguez et al. after archwires replacement[4,28].

Due to the superiority of paracetamol-caffeine (500–65 mg) over acetaminophen alone[12], and a greater level of safety when compared to other locally acting NSAIDs, it was chosen in the current trial as opposed to other investigations[8,9] that solely used paracetamol for orthodontic pain management. Sandue et al. demonstrated that the effectiveness of analgesics depends on the drug's pharmacokinetics in terms of dose and timing of administration[36]. To maintain a healthy drug plasma level in the current study, all patients were instructed to take their prescription an hour before any replacement and to continue taking a pill every 8 h for the first couple of days. Then, as needed, in the following days. This instruction was given as the orthodontic pain peaks after 24–48 h of force application[2,21].

A small amount of spontaneous and chewing pain was perceived directly after each archwire replacement (a mean of 1 to 2 units on the NRS). The compression of the periodontal ligament fibers after orthodontic force application might be responsible for this immediate pain sensation[2,23]. This study showed that most patients experienced the most pain 24 to 48 h after undergoing orthodontic replacement, consistent with the results of many previous reports[2,4,8,16,28]. The time course of orthodontic pain may be related to biological changes after an orthodontic force is applied, specifically, the concentration of interleukin (IL) 1-beta that reaches its highest levels after 24 h[37].

Intergroup findings

Differences less than 2 degrees on the Numeric Rating Scale (NRS) were considered not clinically significant. Understanding the minimally clinically important difference (MCID) is essential for evaluating the effectiveness of pain management modalities[38].

The spontaneous and chewing pain levels induced by orthodontic separators were significantly lower than those from the drug and control groups, just at the peak of pain after 48 h of insertion. These results were similar to Almallah and Qamruddin et al.[21,22]. Qamruddin et al. used 940 nm of GaAlAs laser with 200 mW for 20 s and irradiated at three points: mesial, distal, and the middle of the first molar from the buccal side only. Almallah et al. compared single versus double laser irradiation doses with 4 J/cm^2 energy applied for 28 s on each of the eight selected points (4 buccal and four palatal). Neither the LLLT nor the medication could relieve the orthodontic pain induced by orthodontic separators in the other assessment times. The current results contrasted what was reported by Mirhashemi et al. who evaluated double laser irradiations with 15 J/cm^2 energy applied for 11 s on each point of cervical points for mesial and distal roots of the maxillary 1 st molar[20]. The difference in efficacy of LLLT may be due to variations in overall laser energy, application protocol, or study methodology, as suggested by Farzan et al.[17]. Similarly, the above results match the outcomes of Patel et al., who used only paracetamol one hour before and after 3 and 7 h of separator placement[9]. In contrast, Najafi et al. suggested that preemptive paracetamol alone could be an effective option for managing orthodontic pain; however, their study lacked a control group, and the results should be viewed with care[8].

The pain levels experienced by the participants in the laser, drug, and control groups when the initial archwire (0.014 NITI) was installed were similar to those found in previous studies[7,15]. The 0.014 and 0.016 -NiTi-induced Pain mean scores in LLLT and paracetamol-caffeine groups were similar to those of the control group without significant differences. Conversely, Qamruddin et al. demonstrated that a single dose of Al-Ga-As laser irradiation with a power of 75 J per tooth can reduce the pain associated with the initial archwire[4]. The specific ability of LLLT to reduce the peak pain experienced after the separator placement, rather than after the installation of initial archwires, may be because separators induce significantly greater levels of pain than round archwires, leading to a greater concentration of inflammatory mediators, such as PGE2, IL-1β, and TNF. These mediators create a more suitable environment for laser irradiation to work effectively in reducing pain[1]. The study's results indicate that the combination of paracetamol and caffeine did not provide significant relief for spontaneous and chewing pain caused by initial archwires, similar to what was observed with pain caused by separators. In a previous study by Alshammari et al., it was reported that using paracetamol or chewing gum after initial archwire installation had similar effects[25]. However, their study was limited by the absence of a control group, making it unclear whether the modalities used effectively controlled pain. This is particularly relevant to the current study, as pain mean scores in their study were compared to those of a control group in the current study.

The insertion of rectangular archwires caused mild levels of pain that peaked 24 h after insertion and subsided by the end of the week in all three study groups. The laser in the current study reduced pain levels during its peak at 24 and 48 h after rectangular wire insertion, but the amount of reduction was not clinically important. These results were similar to those reported by Dominguez and Velasquez, who evaluated the effect of a low-energy GaAlAs laser with a wavelength of 830 nm and an energy of 100 J/cm2 for 22 s on the vestibular and palatal/lingual sides after 0.019*0.025 SS archwire insertion in crowded cases that were resolved without extraction[28]. This effect may be attributed to the cumulative effect of successive laser doses applied from the first to the final stage in this study. In the drug group, the mean values of the spontaneous and chewing pain scores after 24 and 48 h of placing the 0.016*0.022 NiTi and 0.019*0.025 NiTi archwires were the lowest among the tested groups. However, the multiple comparisons did not reveal any statistically significant differences between the drug and control groups. The medication used was not effective in alleviating mild pain levels in stages from 0.016*0.022 NITI to 19*25 SS, and this may be attributed to the fact that the pain levels induced in these stages were mild; there was no need to relieve these mild pain levels, and this psychological factor may have affected the drug's efficacy.

The drug intake did not significantly reduce pain levels compared to receiving no treatment, although there was some decrease. Additionally, the addition of caffeine did not appear to enhance the pain-relieving effect of paracetamol on induced pain.

Within group findings

Most previous studies in this area have only examined one stage, either separators[2,8,21,22] or the initial archwire[4,7,15,25], as the source of force. We believe that studies focusing on a single component as the source of force may draw inaccurate conclusions because the nature of pain caused by each stage may have different characteristics compared to the pain caused by other stages. Therefore, to the limit of our knowledge, this study is the first to assess all stages of leveling and alignment treatment with a fixed appliance, providing a comprehensive understanding of orthodontic pain from A to Z.

In the current study, pain levels (spontaneous and chewing) were significantly greater in the separation and initial archwire stages compared to other stages after 24 and 48 h in the laser, drug, and control groups. However, there was no significant difference between these two stages. It is commonly believed that orthodontic pain increases proportionately to the force applied, which may explain why there was more intense pain during initial archwire alignment but decreased with the 0.016-in wire and became almost non-existent later on[4]. This also could be attributed to the patient's exposure to new procedures that could increase anxiety and pain levels[39]. Additionally, patients may become accustomed to the fixed appliance and the pain it causes over time, although there is currently no research to support this theory.

Specifically, the pain levels during separation were slightly greater than those induced by the 0.014 NiTi archwire. This could be interrupted by the fact that the pain caused by the separator differs from that caused by the archwire, as the separator exerts only lateral forces for teeth separation. In contrast, fixed orthodontic treatment using an archwire involves more complex movements, such as extrusion, intrusion, and uncontrolled tipping, which can contribute to a different type of pain[7].

Additionally, no significant differences were detected between the pain mean scores recorded for the initial archwire and 0.016*0.022 NiTi or 0.019*0.025 NiTi. One possible explanation is that the 0.016*0.022 NiTi archwire is the first rectangular archwire that transfers torque data, and the 0.019*0.025 NiTi wire is the largest wire that conveys the majority of torque data through the brackets[40]. Orthodontic pain can be a significant concern following the initial stages of orthodontic treatment, particularly after the placement of separators and the initial archwire installation. However, using buccal tubes can help avoid the need for separators.

Limitations

One of the limitations of the current study was that masking was not applicable due to the long-term follow-up period and patients in the control group did not receive any placebo procedure, and the placebo procedures could play an intervention role in pain studies. Additionally, the current study did not assess participants' pain threshold before enrollment. no drop-outs were planned during sample calculation of this trial. Another limitation of the current study is that the use of medication in the drug group after the first couple of days of treatment was not tracked or analyzed through subgroup or sensitivity analysis.

Conclusions

- LLLT reduced the peak of high-level pain just during separation and was not highly effective throughout the treatment course.
- The paracetamol-caffeine combination was ineffective in reducing pain perception throughout the treatment course.
- It is essential to consider pain management modalities during the initial stages for all patients and throughout the treatment course, just in patients with a low pain threshold.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Author contributions

(AMO): Conception, acquisition, analysis, interpretation of data, drafting of the work, approved the submitted version; (MYH): Conception, acquisition, analysis, interpretation of data, drafting of the work, approved the submitted version, and supervision; (RA): Conception, acquisition, analysis, interpretation of data, drafting of the work, approved the submitted version; (KD), (AB) and (OA): Revision and approved the submitted version; (YL): Writing- Original draft preparation, Methodology, approved the submitted version.

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Competing interests
The authors declare no competing interests.

Ethics approval
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Additional information
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